

Image Processing & Recognition: Exam Solutions

This guide provides step-by-step solutions for the exam problems based strictly on the provided course PDFs.

Part 1: Calculation Problems (Courses 1-5)

Problem 1: Ridge Regression (Course 1)

Statement: Assume we have a dataset

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1, 1), (2, -1), (3, 2)\}$ of points in the 2D plane, and we want to estimate the following ridge regression model with hyperparameter $\lambda = 2$:

$$\hat{y} = w_1 x + w_0$$

Compute w_0 , w_1 and $\hat{y}^{(1)}, \hat{y}^{(2)}, \hat{y}^{(3)}$.

Solution:

1. Formulate the Matrices From **Course 1, Slide 20**, we construct the design matrix X and target vector y . Since the model has a bias term w_0 , we add a column of 1s to X .

- $n = 3$ (number of samples)
- $d = 1$ (number of features)

$$X = \begin{bmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ 1 & x^{(3)} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

2. Apply the Ridge Regression Formula From **Course 1, Slide 66**, the closed-form solution for the weights w^* in ridge regression is:

$$w^* = (X^T X + \lambda I)^{-1} X^T y$$

Step 2a: Compute $X^T X$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1+1+1 & 1+2+3 \\ 1+2+3 & 1+4+9 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

Step 2b: Add the Regularization Term (λI) Given $\lambda = 2$:

$$\lambda I = 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(X^T X + \lambda I) = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 6 & 16 \end{bmatrix}$$

Step 2c: Compute the Inverse For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Determinant: $(5 \times 16) - (6 \times 6) = 80 - 36 = 44$.

$$(X^T X + \lambda I)^{-1} = \frac{1}{44} \begin{bmatrix} 16 & -6 \\ -6 & 5 \end{bmatrix}$$

Step 2d: Compute $X^T y$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1(1) + 1(-1) + 1(2) \\ 1(1) + 2(-1) + 3(2) \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Step 2e: Compute Final Weights w^*

$$w^* = \frac{1}{44} \begin{bmatrix} 16 & -6 \\ -6 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 16(2) + (-6)(5) \\ -6(2) + 5(5) \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 32 - 30 \\ -12 + 25 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 2 \\ 13 \end{bmatrix}$$

$$w_0 = \frac{2}{44} \approx 0.045, \quad w_1 = \frac{13}{44} \approx 0.295$$

3. Compute Predictions ($\hat{y}$) From **Course 1, Slide 20**, $\hat{y} = Xw$.

$$\hat{y} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2/44 \\ 13/44 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 1(2) + 1(13) \\ 1(2) + 2(13) \\ 1(2) + 3(13) \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 15 \\ 28 \\ 41 \end{bmatrix}$$

Answer: $w_0 = 2/44 \approx 0.045$ $w_1 = 13/44 \approx 0.295$ $\hat{y}^{(1)} = 15/44 \approx 0.34$ $\hat{y}^{(2)} = 28/44 \approx 0.64$
 $\hat{y}^{(3)} = 41/44 \approx 0.93$

Problem 2: MLP Regression (Course 2)

Statement: Draw the diagram of an MLP for regression, having 2 inputs, two hidden layers of 2 and 3 units, respectively, and 3 outputs. The activation function for both hidden layers is ReLU. For the input $X = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, if the weight matrices and bias vectors of the layers are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the output O of the network. Specify the total number of real parameters (weights and biases) for the network.

Solution:

1. Diagram Description Based on Course 2, Slide 23:

- **Input Layer:** 2 nodes (x_1, x_2)
- **Hidden Layer 1:** 2 nodes ($h_1^{(1)}, h_2^{(1)}$)
- **Hidden Layer 2:** 3 nodes ($h_1^{(2)}, h_2^{(2)}, h_3^{(2)}$)
- **Output Layer:** 3 nodes (o_1, o_2, o_3)

2. Parameter Generation (Random Selection from $\{-1, 0, 1\}$)

- **Layer 1 ($W^{(1)}, b^{(1)}$):** Input size 2, Output size 2.

$$W^{(1)} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

- **Layer 2 ($W^{(2)}, b^{(2)}$):** Input size 2, Output size 3.

$$W^{(2)} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} 0 & -1 & 1 \end{bmatrix}$$

- **Layer 3 ($W^{(3)}, b^{(3)}$):** Input size 3, Output size 3.

$$W^{(3)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad b^{(3)} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

3. Forward Propagation Formulas from **Course 2, Slides 25 & 31:** $H = \text{ReLU}(XW + b)$ and $O = HW + b$.

- **Hidden Layer 1:**

$$\begin{aligned} Z^{(1)} &= XW^{(1)} + b^{(1)} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

$$H^{(1)} = \text{ReLU}(Z^{(1)}) = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}$$

- **Hidden Layer 2:**

$$Z^{(2)} = H^{(1)}W^{(2)} + b^{(2)} = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ 2 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$H^{(2)} = \text{ReLU}(Z^{(2)}) = \begin{bmatrix} 2 & 0 & 3 \\ 2 & 0 & 3 \end{bmatrix}$$

- **Output Layer:**

$$O = H^{(2)}W^{(3)} + b^{(3)} = \begin{bmatrix} 2 & 0 & 3 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 2 & 0 & 3 \end{bmatrix}$$

4. Total Parameters

- Layer 1: $(2 \text{ inputs} \times 2 \text{ units}) + 2 \text{ biases} = 6$
- Layer 2: $(2 \text{ inputs} \times 3 \text{ units}) + 3 \text{ biases} = 9$
- Layer 3: $(3 \text{ inputs} \times 3 \text{ units}) + 3 \text{ biases} = 12$
- **Total:** $6 + 9 + 12 = 27$

Problem 3: Convolutional Layer Calculations (Course 3)

Statement: Assume we have an input image whose shape is $3 \times 224 \times 224$. Compute the output shape and number of parameters (weights and biases) of each of the following possible settings for the first convolutional layer, given the input has the shape described above. (a) The first convolutional layer has a stride of 2, a kernel size of 3, input padding of 0, and 64 output channels. (b) The first convolutional layer has a stride of 4, a kernel size of 5, input padding of 2, and 32 output channels.

Solution:

Formulas (Course 3):

- **Output Shape:** $\lfloor (n_{in} - k + 2p) / s \rfloor + 1$ (Slide 46).
- **Parameters:** $(k \times k \times c_{in} + 1) \times c_{out}$ (Slide 55 concept extended to standard conv).

Part (a):

- Input: 224×224 , Channels (c_{in}) = 3.

- Hyperparameters: $s = 2, k = 3, p = 0, c_{out} = 64$.
- **Output Height/Width:** $\lfloor (224 - 3 + 2(0))/2 \rfloor + 1 = \lfloor 221/2 \rfloor + 1 = 110 + 1 = 111$.
- **Output Shape:** $64 \times 111 \times 111$.
- **Parameters:** $(3 \times 3 \times 3 + 1) \times 64 = (27 + 1) \times 64 = 28 \times 64 = 1792$.

Part (b):

- Input: 224×224 , Channels (c_{in}) = 3.
- Hyperparameters: $s = 4, k = 5, p = 2, c_{out} = 32$.
- **Output Height/Width:**
 $\lfloor (224 - 5 + 2(2))/4 \rfloor + 1 = \lfloor (224 - 5 + 4)/4 \rfloor + 1 = \lfloor 223/4 \rfloor + 1 = 55 + 1 = 56$.
- **Output Shape:** $32 \times 56 \times 56$.
- **Parameters:** $(5 \times 5 \times 3 + 1) \times 32 = (75 + 1) \times 32 = 76 \times 32 = 2432$.

Problem 4: RNN (Course 4)

Statement: Draw the diagram of an RNN for regression, having 3 inputs, a hidden state of dimension 2, and 3 outputs. The activation function for the hidden state is ReLU. For the inputs

$$X_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, X_2 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix},$$

if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs O_1 and O_2 of the network. Specify the total number of real parameters (weights and biases) for the network.

Solution:

1. Diagram Based on **Course 4, Figure 3 (Slide 31)**: Input X feeds into Hidden H , which feeds into Output O . There is a recurrent arrow from H_t to H_{t+1} .

2. Parameter Generation (Random)

- Inputs $d = 3$, Hidden $h = 2$, Outputs $q = 3$.
- **Input-to-Hidden (W_{xh}):** 3×2 .

$$W_{xh} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

- **Hidden-to-Hidden (W_{hh}):** 2×2 .

$$W_{hh} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- **Hidden-to-Output (W_{hq}):** 2×3 .

$$W_{hq} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

- **Biases:** $b_h = [0, 0]$, $b_q = [0, 0, 0]$.
- **Initial Hidden State (H_0):** 2×2 zeros.

3. Calculations Step 1 ($t = 1$): Formulas (Course 4, Slide 26 & 28):

$$H_t = \phi(X_t W_{xh} + H_{t-1} W_{hh} + b_h), O_t = H_t W_{hq} + b_q.$$

- $X_1 W_{xh} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$
- $H_0 W_{hh} = \mathbf{0}$.
- $H_1 = \text{ReLU} \left(\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- $O_1 = H_1 W_{hq} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$

Step 2 ($t = 2$):

- $X_2 W_{xh} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$
- $H_1 W_{hh} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- Sum: $\begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix}$
- $H_2 = \text{ReLU}(\text{Sum}) = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$
- $O_2 = H_2 W_{hq} = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 0 & -3 \end{bmatrix}$

4. Total Parameters

- $W_{xh} : 3 \times 2 = 6$
- $W_{hh} : 2 \times 2 = 4$
- $b_h : 2$
- $W_{hq} : 2 \times 3 = 6$
- $b_q : 3$
- **Total:** $6 + 4 + 2 + 6 + 3 = 21$.

Problem 5: Attention Mechanism (Course 5)

Statement: Given the 4 key-value pairs

$\{(k_1, v_1), (k_2, v_2), (k_3, v_3), (k_4, v_4)\} = \{([2, 0]^T, [-1, 1, 0]^T), ([0, 1]^T, [-1, 0, 0]^T), ([0, 2]^T, [0, 1, -1]^T), ([1, 1]^T, [1, 0, 1]^T)\}$, and the query $q = [1, 2]^T$, compute the output of attention pooling for scaled dot-product attention.

Solution:**1. Formula From Course 5, Slide 22:**

$$a(q, k) = \frac{q^T k}{\sqrt{d}}$$

Here, q and k are 2D vectors, so $d = 2$. Scaling factor is $\sqrt{2}$.

2. Calculate Scores

- $q^T k_1 = [1, 2] \cdot [2, 0] = 2$. Scaled: $2/\sqrt{2} = \sqrt{2}$.
- $q^T k_2 = [1, 2] \cdot [0, 1] = 2$. Scaled: $2/\sqrt{2} = \sqrt{2}$.
- $q^T k_3 = [1, 2] \cdot [0, 2] = 4$. Scaled: $4/\sqrt{2} = 2\sqrt{2}$.
- $q^T k_4 = [1, 2] \cdot [-1, 0] = -1$. Scaled: $-1/\sqrt{2} = -0.707$.

3. Calculate Weights (Softmax)

$$sum_exp = e^{\sqrt{2}} + e^{\sqrt{2}} + e^{2\sqrt{2}} + e^{-0.707} \approx 4.11 + 4.11 + 16.91 + 0.49 = 25.62$$

- $\alpha_1 = 4.11/25.62 \approx 0.16$
- $\alpha_2 = 4.11/25.62 \approx 0.16$
- $\alpha_3 = 16.91/25.62 \approx 0.66$
- $\alpha_4 = 0.49/25.62 \approx 0.02$

4. Compute Output

$$\begin{aligned} Output &= \sum \alpha_i v_i \\ &= 0.16 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + 0.16 \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + 0.66 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + 0.02 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -0.16 \\ 0.16 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.16 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.66 \\ -0.66 \end{bmatrix} + \begin{bmatrix} 0.02 \\ 0.04 \\ 0.02 \end{bmatrix} \\ &= \begin{bmatrix} -0.16 - 0.16 + 0 + 0.02 \\ 0.16 + 0 + 0.66 + 0.04 \\ 0 + 0 - 0.66 + 0.02 \end{bmatrix} = \begin{bmatrix} -0.30 \\ 0.86 \\ -0.64 \end{bmatrix} \end{aligned}$$

Part 2: Architecture Specification Guide (Problems 6-10)

For these problems, you are not calculating numbers. You are writing a "recipe" for a neural network using the notation provided in the exam instructions:

- FC N (Fully Connected)
- CONV N, K, S, P, D (Convolution)
- POOL K, S (Pooling)

Problem 6: SSD (Course 6)

Task: Define the SSD model architecture (Figure 4, Lecture 6) for an input of $1 \times 3 \times 256 \times 256$ to detect 80 classes.

How to Solve:

1. **Base Network:** SSD starts with a standard CNN (usually VGG-16). You need to write out the layers of VGG-16 (Conv $\rightarrow$ Relu $\rightarrow$ Pool).
2. **Multiscale Feature Maps:** After the base, add extra Convolutional layers that reduce the size (e.g., $10 \times 10 \rightarrow 5 \times 5 \rightarrow 3 \times 3 \rightarrow 1 \times 1$).
3. **Prediction Heads:** For *each* of these feature map scales, you attach a separate Convolutional layer to predict:
 - **Classes:** $N_{anchors} \times (Classes + 1)$
 - **Boxes:** $N_{anchors} \times 4$
4. **Reference:** Course 6, Figure 4 (Slide 58).

Problem 7: DeepLab-ASPP (Course 7)

Task: Define DeepLab-ASPP (Figure 15b, Lecture 7) with VGG-16 backbone for input $1 \times 3 \times 256 \times 256$ and 20 classes.

How to Solve:

1. **Backbone:** Write the VGG-16 layers. **Crucial Detail:** In DeepLab, the last pooling layers often have `stride=1` (instead of 2) to keep the resolution high, using `dilation` in subsequent convs instead.
2. **ASPP Module:** This is the key. You need to write 4 parallel CONV layers, each taking the *same* input but having different `dilation (D)` values (e.g., 6, 12, 18, 24).
3. **Fusion:** Concatenate the outputs of these 4 layers.
4. **Classifier:** Add a 1×1 convolution to map to 21 classes (20 + background).
5. **Reference:** Course 7, Figure 15b (Slide 54).

Problem 8: Visual Dialog - LF Encoder (Course 8)

Task: Define the Late Fusion (LF) Encoder (Figure 12, Lecture 8). Input is $1 \times 3 \times 256 \times 256$ image.

How to Solve:

1. **Image Stream:** Write a CNN (like VGG) that ends with a Flatten or FC layer to get a vector (e.g., size 4096).
2. **Question Stream:** Write an LSTM layer that takes word embeddings and outputs a vector (e.g., size 512).
3. **History Stream:** Write another LSTM layer for the dialog history (e.g., size 512).
4. **Late Fusion:** Concatenate these three vectors.
5. **Decoder:** Add FC layers to predict the answer.
6. **Reference: Course 8, Figure 12 (Slide 52).**

Problem 9: 3D CNN (Course 9)

Task: Define C3D (Figure 8, Lecture 9). Input video is $1 \times 3 \times 16 \times 256 \times 256$ (16 frames).

How to Solve:

1. **Notation Change:** The prompt uses CONV N, K, S, P, D. For 3D CNN, assume K, S, P refer to (K_t, K_h, K_w) etc.
2. **Structure:** Follow C3D exactly.
 - CONV 64, 3, 1, 1, 1 (3D Conv).
 - POOL 2, 2 (3D Max Pool - reduces time, height, and width by 2).
 - Repeat this pattern (Conv -> Pool) 5 times.
 - End with two FC 4096 layers and one FC 101 (for 101 classes).
3. **Reference: Course 9, Figure 8 (Slide 39).**

Problem 10: DCGAN (Course 10)

Task: Define DCGAN (Figure 2, Lecture 10) to generate 128×128 images.

How to Solve:

1. **Generator (Input: Noise Vector):**
 - Start with a FC layer to reshape noise into a small tensor (e.g., $4 \times 4 \times 1024$).
 - Use **Transposed Convolutions** (CONVTR) to double the size repeatedly ($4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128$).
 - You need enough layers to reach 128.